
Tomographic Reconstruction of Expansion-Dominated Spacetime Using Null Geodesic Sampling

A Multi-Line-of-Sight Inversion Framework for Curvature-Minimum Geometry

James J. Coit

Abstract

Mass Deficit Nodes (MDNs) have been identified as curvature-minimum spacetime structures organizing expansion-dominated regions of the cosmic web. Previous work established that photon propagation through MDN-organized geometry produces a non-canceling, path-integrated contribution to cosmological redshift. Paper 5 introduced a depth-slicing method for isolating geometric contributions along individual lines of sight, and Paper 6 introduced a complementary calibration framework using the cosmic microwave background (CMB) as a dual null probe.

In this work, we synthesize these developments into a **tomographic reconstruction framework** for inferring the underlying spacetime expansion field from observational data. By treating redshift differences across multiple lines of sight as path-integrated measurements of an effective expansion field, we formulate an inverse problem for recovering the spatial structure of expansion-dominated geometry. We incorporate geometric coherence constraints, observational noise, and optional CMB calibration into a unified framework.

This establishes a new observational program: **null geodesic tomography of spacetime structure**, enabling direct reconstruction of curvature-minimum geometry using existing datasets.

1. Introduction

Cosmological observations fundamentally probe spacetime through null propagation. In previous papers, we demonstrated that in an inhomogeneous universe organized by curvature extrema,

photon trajectories accumulate a **path-integrated geometric contribution to redshift** arising from differences between null and timelike congruence evolution.

Paper 5 introduced a method to isolate this contribution within finite depth intervals through differential line-of-sight comparisons. Paper 6 established that independent null observables, such as the cosmic microwave background, encode consistent information about the same intervening geometry.

The present work extends these results by addressing the following question:

Can spacetime geometry itself be reconstructed from null geodesic observables?

We answer this affirmatively by formulating the problem as an inverse reconstruction of an effective expansion field sampled along multiple null trajectories.

2. Forward Model: Null-Sampled Expansion Field

From prior results, the logarithmic redshift along a null trajectory is given by:

$$\ln(1+z) = \frac{1}{3} \int_{\lambda_e}^{\lambda_o} \Theta_{\text{eff}}(x^\mu(\lambda)) d\lambda$$

where:

- $\Theta_{\text{eff}}(x)$ is the effective expansion scalar sampled by null geodesics
- λ is the affine parameter along the trajectory

2.1 Depth-Sliced Form

For differential measurements between two sources:

$$\delta z_i \approx \frac{1}{3} \int_{\text{shell}_i} \Theta_{\text{eff}}(x) d\lambda + \epsilon_i$$

where:

- i indexes pairs of sources
- ϵ_i includes observational noise

3. Discrete Inverse Formulation

We discretize spacetime into voxels labeled by index j , each with constant expansion value θ_j .

Each line of sight defines a set of path-length overlaps L_{ij} .

The observable becomes:

$$\delta z_i = \sum_j K_{ij} \theta_j + \epsilon_i$$

where:

$$K_{ij} = \frac{1}{3} L_{ij}$$

3.1 Interpretation

- δz_i : observable data vector
- θ_j : unknown expansion field
- K_{ij} : geometric sampling matrix

3.2 Inverse Problem

We seek:

$$\theta_j = \operatorname{argmin}[\| \mathbf{K}\theta - \delta z \|^2 + \mathcal{R}(\theta)]$$

where $\mathcal{R}$ is a regularization term.

4. Geometric Sampling Structure

Each null trajectory samples spacetime along a 1D path.

With multiple sources:

- different angles $\rightarrow$ lateral sampling
- different depths $\rightarrow$ radial sampling

This produces a 3D sampling network.

4.1 Coherence Constraint

From Paper 5:

$$\Delta x(r) = r \theta_{\text{sep}}$$

Coherent sampling requires:

$$r \theta_{\text{sep}} \ll R_{\text{void}}$$

Define:

$$r_{\text{coh}} = \frac{R_{\text{void}}}{\theta_{\text{sep}}}$$

4.2 Coherence Kernel

Sampling weight:

$$W_{ij} = \exp \left[-\frac{r_{ij}^2 \theta_i^2}{2R_{\text{void}}^2} \right]$$

Effective matrix:

$$K_{ij}^{\text{eff}} = W_{ij} K_{ij}$$

5. Noise Model

Observational residuals may be decomposed as:

$$\epsilon_i = \epsilon_{\text{pec},i} + \epsilon_{\text{meas},i} + \epsilon_{\text{coh},i}$$

Components:

- peculiar velocity differences
 - redshift measurement uncertainty
 - partial coherence breakdown
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5.1 Statistical Properties

To leading order:

$$\begin{aligned}\langle \epsilon_i \rangle &= 0 \\ \langle \epsilon_i \epsilon_j \rangle &\approx \sigma_i^2 \delta_{ij}\end{aligned}$$

6. Incorporation of CMB Calibration

From Paper 6:

$$\Delta z_{\text{MDN}} \approx F \cdot \left(\frac{\Delta T}{T} \right)$$

This provides an independent estimate of:

$$\sum_j K_{ij} \Theta_j$$

6.1 Joint Constraint

We augment the system:

$$\begin{bmatrix} \delta z \\ \left(\frac{\Delta T}{T}\right) \end{bmatrix} = \begin{bmatrix} \mathbf{K} \\ \mathbf{C} \end{bmatrix} \Theta + \text{noise}$$

7. Reconstruction Strategy

Several inversion strategies may be used:

7.1 Least Squares

$$\Theta = (\mathbf{K}^T \mathbf{K})^{-1} \mathbf{K}^T \delta z$$

7.2 Regularized Solution

$$\Theta = (\mathbf{K}^T \mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{K}^T \delta z$$

7.3 Bayesian Inference

$$P(\Theta \mid \delta z) \propto P(\delta z \mid \Theta) P(\Theta)$$

8. Resolution and Sampling Limits

Resolution depends on:

- angular density of sources
 - redshift coverage
 - coherence scale
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8.1 Effective Resolution Scale

$$\Delta x_{\text{res}} \sim \min \left(\frac{1}{n_{\text{src}}^{1/3}}, R_{\text{void}} \right)$$

9. Physical Interpretation of Reconstructed Field

The reconstructed field $\Theta(x)$ represents:

- expansion-dominated regions (void interiors)
 - transition zones near filaments
 - spatial variation of null-congruence divergence
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10. Observational Implementation

Using existing datasets:

- Pantheon+ supernovae
- SDSS galaxy redshifts
- Planck CMB maps

One can:

1. construct δz_i
 2. apply selection function
 3. build sampling matrix K_{ij}
 4. perform inversion
 5. compare with known structure
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11. Testable Predictions

11.1 Recovered Void Structure

Reconstruction should correlate with known void catalogs.

11.2 Anisotropic Expansion Features

Non-uniform $\Theta(x)$ predicted.

11.3 Consistency Across Observables

Agreement between:

- redshift-based inference
 - CMB calibration
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12. Limitations

- sparse angular sampling at high redshift
 - noise from peculiar velocities
 - approximate treatment of null geodesics
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13. Discussion

The framework developed here represents a shift from:

- parameter inference

to:

- direct geometric reconstruction

Spacetime is treated not as an assumed background, but as an **observable structure inferred from null propagation**.

14. Conclusion

We have formulated a tomographic reconstruction framework for inferring expansion-dominated spacetime structure using null geodesic sampling.

This method:

- builds on depth-sliced observables
- incorporates geometric coherence constraints
- integrates dual null probes
- enables inversion of path-integrated redshift measurements

It establishes a new observational paradigm:

Null Geodesic Tomography of Spacetime